

# Strings Practice Sheet

## Easy

### 1. Input and Output of Strings

Take a string input and print it using getline (for C++).

### 2. String Length

Write a program to find the length of a string without using built-in functions.

### 3. Count Vowels and Consonants

Count the number of vowels and consonants in a given string.

### 4. Reverse a String

Write a program to reverse a string using a loop.

### 5. Change Case

Convert all lowercase characters to uppercase and vice versa.

### 6. Check Palindrome

Check whether a string is a palindrome (same forward and backward).

## Medium

### 1. Count Words

Count the number of words in a sentence (words are separated by spaces).

### 2. Remove Spaces

Remove all white spaces from a given string.

### 3. Character Frequency

Print the frequency of each character in a string.

### 4. Compare Two Strings

Write a custom strcmp function to compare two strings without using built-in functions.

### 5. Check Anagram

Check whether two strings are anagrams of each other.

### 6. Sort Characters

Sort all characters of a string in alphabetical order.

### 7. Replace Substring

Find and replace all occurrences of a given substring in the string.

## Hard

### 1. Longest Word in Sentence

Find the longest word in a given sentence.

## 2. Longest Palindromic Substring

Find the longest palindromic substring in a given string.

## 3. Remove Duplicates

Remove all duplicate characters from a string.

## 4. First Non-Repeating Character

Find the first non-repeating character in a string.

## 5. Substring Search

Implement a function to check if one string is a substring of another.

## 6. String Compression

Compress a string such as aabcccccaaa to a2b1c5a3.

## 7. ZigZag Conversion

Convert a string into zigzag pattern and return the result row-wise.